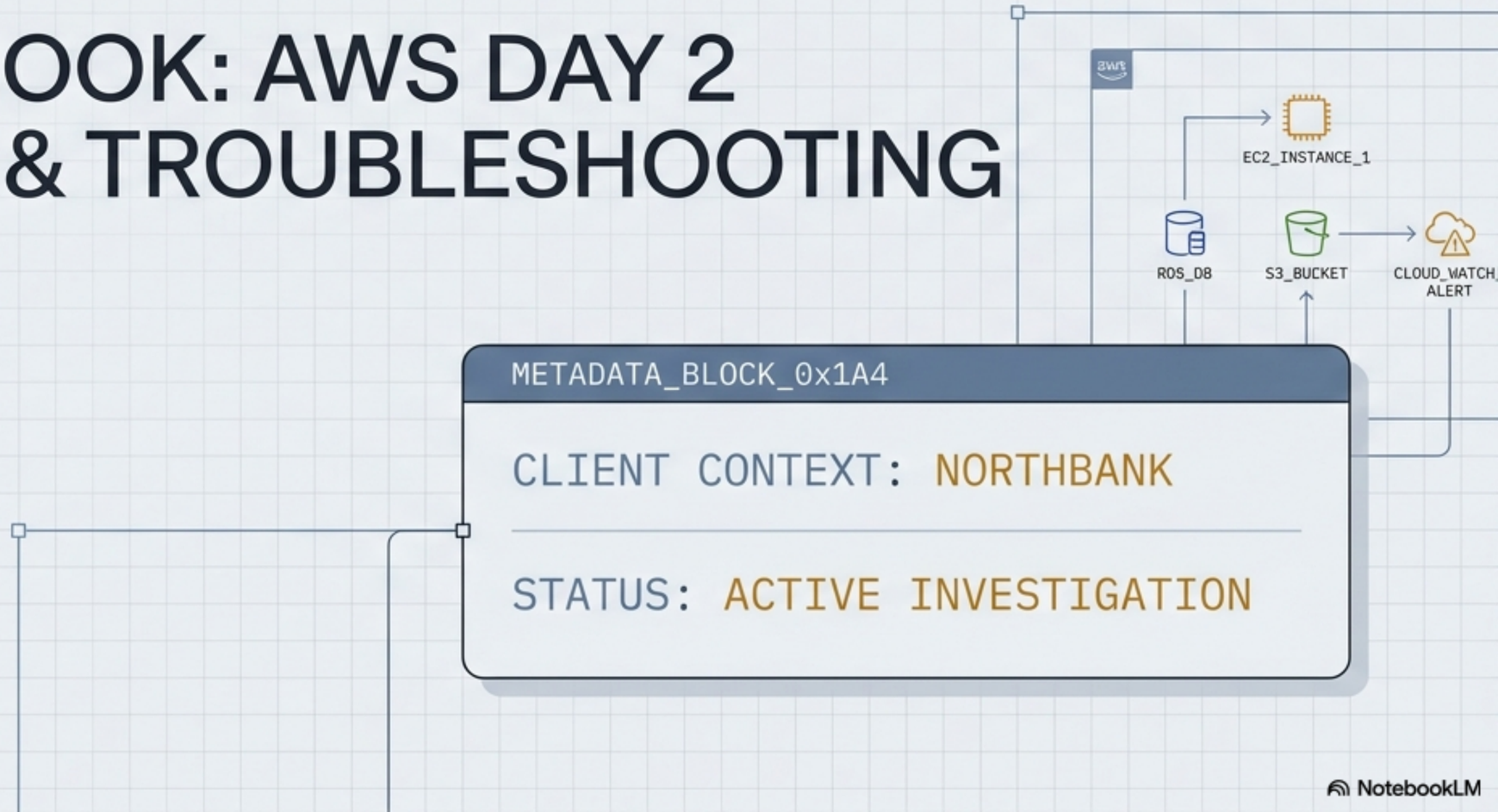
INCIDENT REPORT: THE MONTHLY COST SPIKE

LAB 14 PLAYBOOK: AWS DAY 2 OPERATIONS & TROUBLESHOOTING



NotebookLM

Reporter: Tom (Finance)

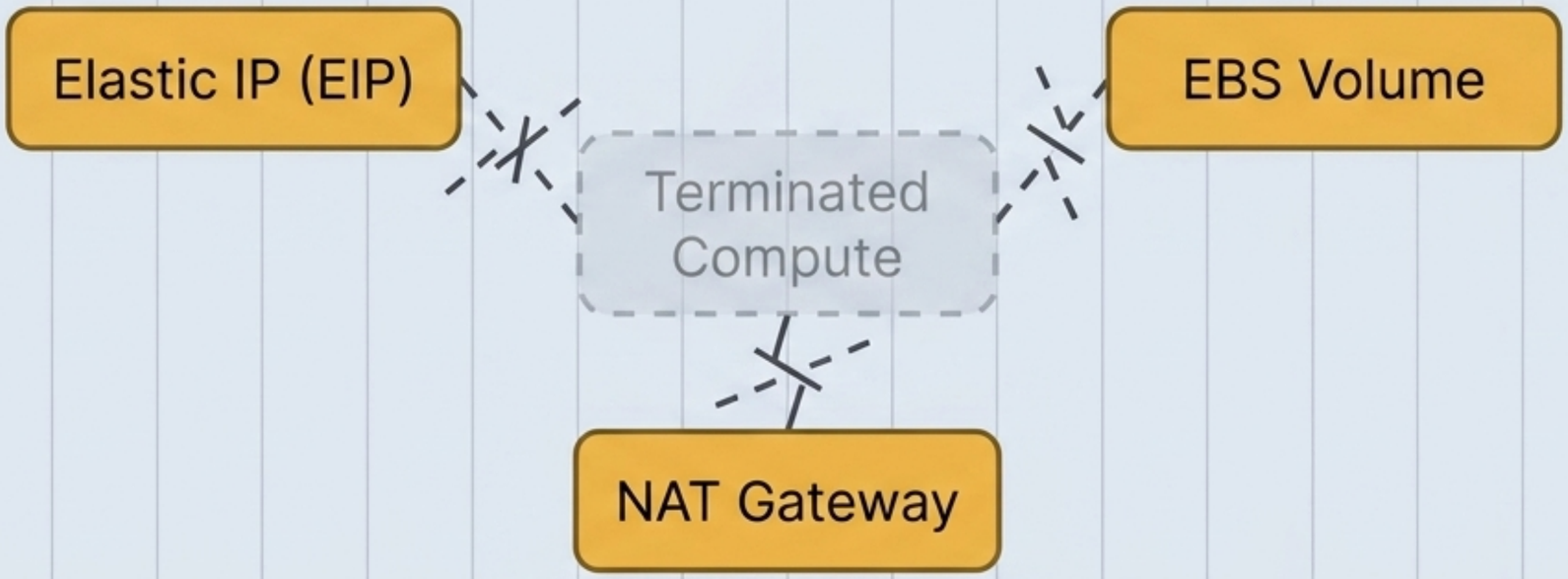
Subject: The bill jumped — find out why.

Message: The AWS bill went up significantly this month. We need to find what's driving it immediately.

```
Monthly Task trigger received. Initiating triage protocol.
```

NotebookLM

The Usual Suspects: Idle and Unattached



Root Cause Hypothesis: Resources billed independently of their parent compute instances.

Key Insight: Idle and unattached resources are the #1 driver of surprise cloud costs.

NotebookLM

Triage Tooling: Cost Explorer vs. Direct Queries

	AWS Cost Explorer	Direct CLI Queries
Data Latency	~24-hour lag	Real-time state
Primary Use Case	Trend analysis & financial reporting	Fast triage & active incident response
Target Audience	Finance/Management	DevOps/Engineering

Standard operating procedure for active cost spikes: Bypass Cost Explorer. Query the resources directly.

NotebookLM

Environment Setup: Simulating the Leak

```
$ cd labs/14-cost

(Note: no exports needed)

$ ./break.sh
```



STATE: VULNERABLE (Bleeding funds)

NotebookLM

Phase 1: Detect the Anomalies

Unattached Elastic IPs

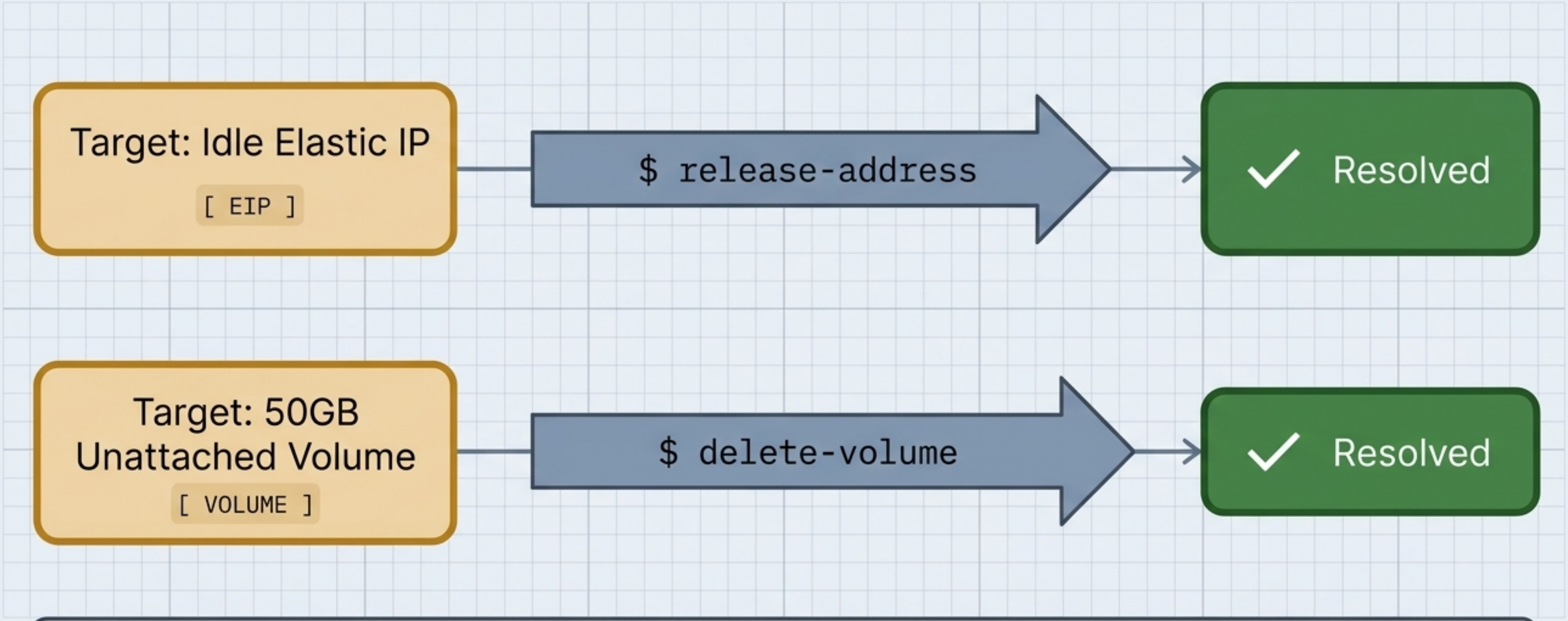
```
$ describe-addresses
{
    "Addresses": [
        {
            "PublicIp": "54.21.231.154",
            "AllocationId": "eipalloc-0d6a6b2d3b1e5a8f2",
            "Domain": "vpc",
            "NetworkInterfaceId": null,
            "AssociationId": null,
            "PrivateIpAddress": null
        }
```

Unattached EBS Volumes

```
$ describe-volumes
{
    "Volumes": [
        {
            "VolumeId": "vol-0a1b2c3d4e5f6g7h8",
            "Size": 50,
            "VolumeType": "gp3",
            "status": "available",
            "attachments": [],
            "CreateTime": "2024-05-20T14:30:00.000Z",
            "AvailabilityZone": "us-east-1a"
        }
    ]
```

NotebookLM

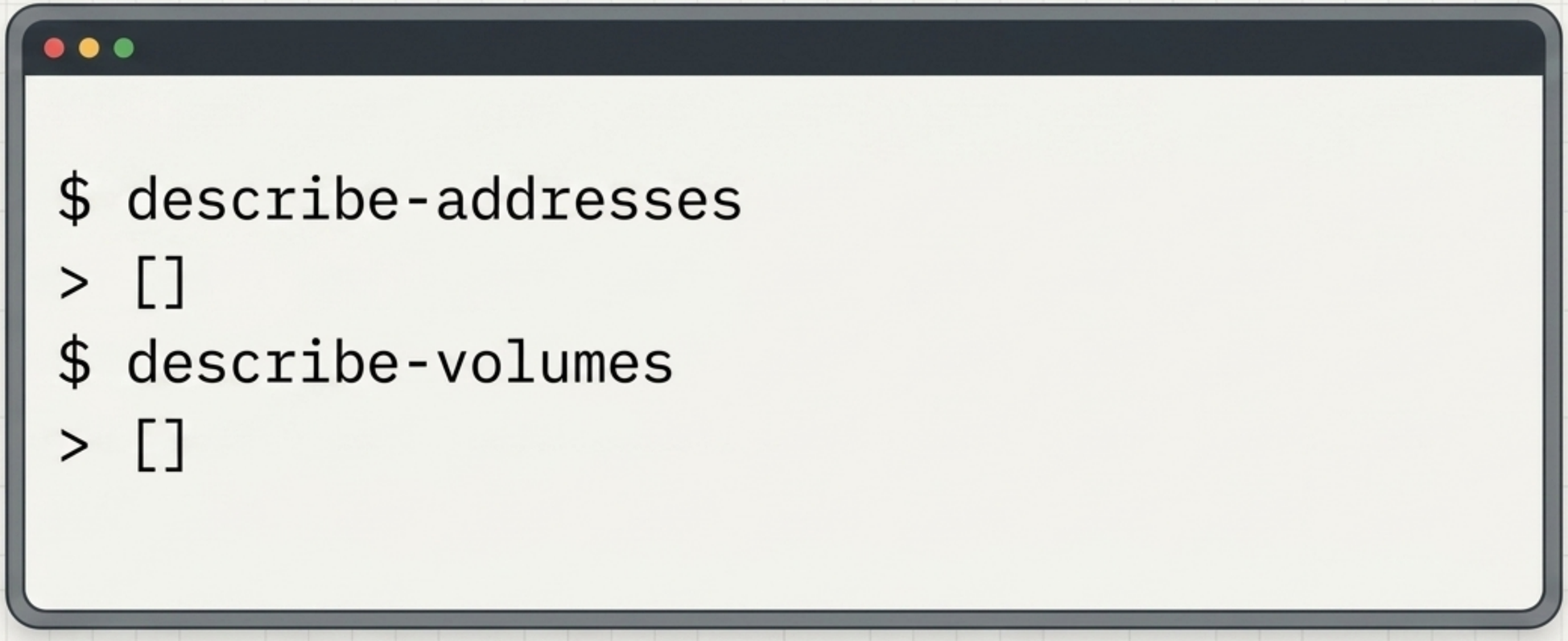
Phase 2: Terminate the Waste



Operational Note: Execution immediately stops the ongoing hourly billing for these specific resources.

NotebookLM

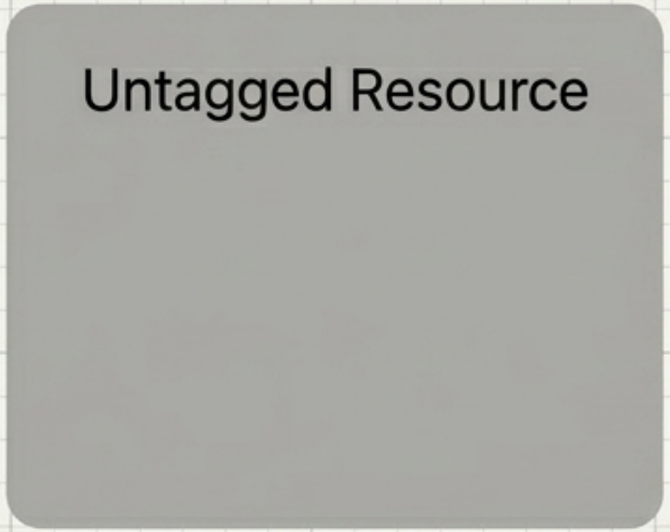
Phase 3: Verify the Clean-Up



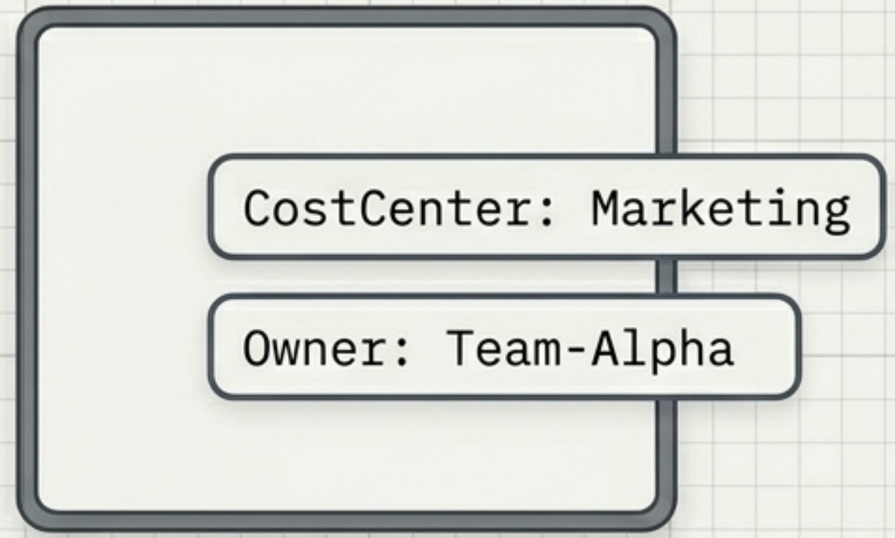
System State: CLEAN. Zero unattached assets detected.

NotebookLM

Day-2 Strategy: The Tagging Prerequisite



Anonymous waste. Impossible to attribute to a team or project.



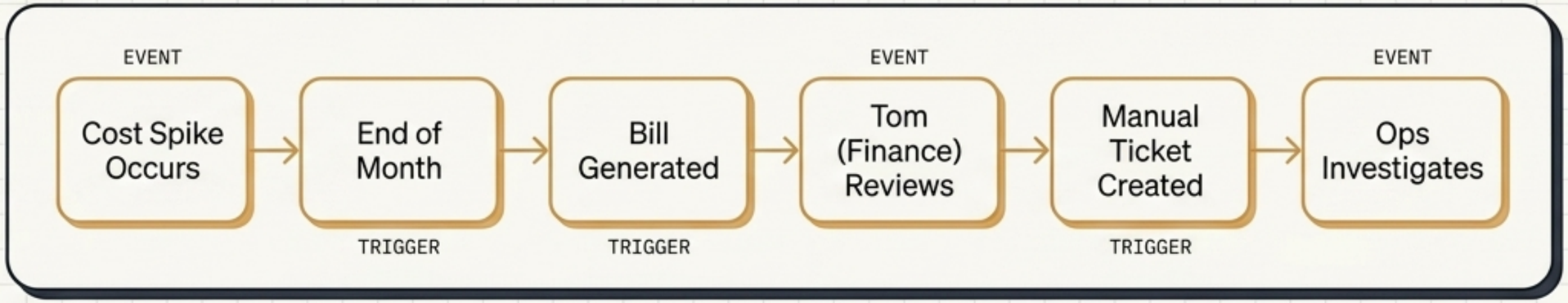
Tagging is what makes cost attribution possible at all.

Core Insight: Without mandatory tagging policies, resolving Tom's ticket requires a system-wide scavenger hunt instead of a targeted inquiry.

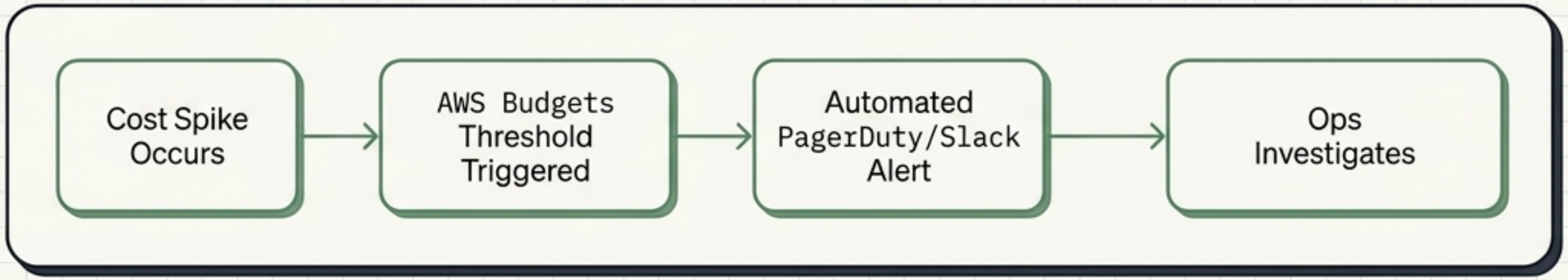
NotebookLM

Day-2 Strategy: Intercepting the Spike

Slow, reactive, high friction



Fast, proactive, intercepts before Finance sees it



Set up budgets so a spike pages you, not Finance.

NotebookLM

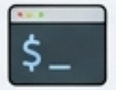
SOP: Rapid Cost Triage

1. Assume the Usual Suspects



Look for idle EIPs, unattached EBS volumes, and forgotten NAT gateways gateways first.

2. Bypass the GUI



Use CLI (describe-addresses, describe-volumes) for real-time data instead of waiting on Cost Explorer's 24h lag.

3. Purge



Execute release-address and delete-volume to stop the bleeding.

4. Automate Prevention



Enforce resource tagging for attribution and configure AWS Budgets to alert Engineering instantly.

NotebookLM